Robustness across conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Robustness across conditions

Compare frozen procedures across declared conditions without replacing slices with one pooled score.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of robustness across conditions

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of inspecting failures and anomalies

Reference example, not the only acceptable wording: Teacher explanation: Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of reproducible environment and records

Reference example, not the only acceptable wording: Teacher explanation: A reproducible environment record links code, data, configuration and runtime dependencies to results. A seed alone is insufficient.

Record exact data IDs/splits, code version, dependency versions, parameters, randomness policy and raw **output**s. Verify hashes for integrity without treating them as correctness proofs.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of predefining an analysis plan

Reference example, not the only acceptable wording: Teacher explanation: An analysis plan records outcomes, units, exclusions, comparisons and uncertainty methods before primary results are inspected.

Deviations remain visible. Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting. Timestamp/version the plan; label later choices exploratory.

Pictured end goal

Explain robustness across conditions using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Inspecting failures and anomalies

Reproducible environment and records

Predefining an analysis plan

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Robustness check**s examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of robustness across conditions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Robustness checks examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause.

Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend **evidence** but cannot guarantee all conditions.

Robustness item	Complete supplied exercise contract
Main condition counts	Current sources: 9/10 pass; outdated sources: 5/10
Main interpretation	Condition-associated descriptive difference; no causal mechanism identified
Separate picture	Frozen classifier and rubric unchanged; indoor 7/8 correct, outdoor 3/8
Picture pooled count	Ten correct among sixteen cases
Needed for a controlled follow-up	Aligned baseline, scoring and eligibility; recorded case/condition changes
Not supplied	Per-case outputs, matched-case transition identities, stable population rates or significance

Robustness item	Complete supplied exercise contract
Scope	Fictional supplied summaries; no new model run or data collection

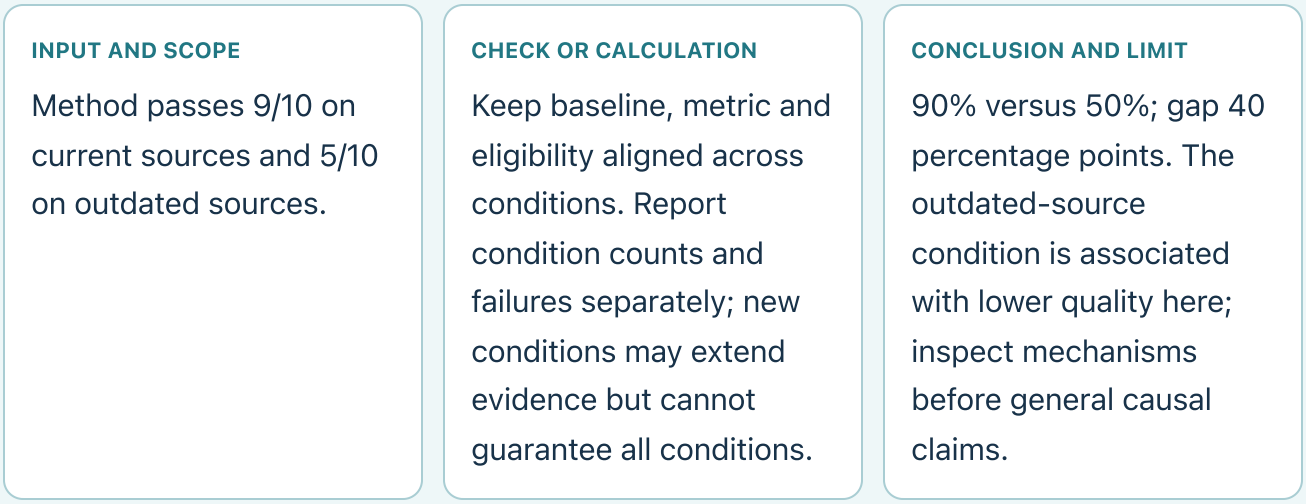
Main evidence ledger and checked key

Main quantity	Current sources	Outdated sources
Passes / evaluated	9/10	5/10
Observed rate	90%	50%
Current minus outdated	40 percentage points	Lower quality associated with this condition here
Mechanism	Not identified	Inspect source validity and procedure traces
Generalization	Only supplied cases	No universal causal or robustness conclusion

Main current 9/10 = 90%, outdated 5/10 = 50%, gap forty percentage points. Do not claim the source condition alone caused the difference. Picture indoor 7/8 = 87.5%, outdoor 3/8 = 37.5%, gap fifty points; pooled 10/16 = 62.5% hides the weaker slice. Independent condition A baseline 6/8 = 75% and candidate 7/8 = 87.5% gives +12.5 points; condition B baseline 5/8 = 62.5% and candidate 4/8 = 50% gives –12.5 points.

Both pooled totals are 11/16, so the tie hides opposite condition effects. Per-case flip identities are not supplied despite paired-rate wording. No significance, universal advantage or causal attribution follows.

Robustness across conditions — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims. Qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Method passes 9/10 on current sources and 5/10 on outdated sources.

90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Robustness audit	Checked interpretation
Overall score looks acceptable	A weak relevant condition can still be hidden
Same frozen model and rubric	Useful declared controls, not proof all case properties align
Outdoor score lower	Descriptive pattern, not a root-cause diagnosis
New condition tested	Extends evidence to that condition only
One candidate improves one slice	Check whether another slice worsens
Two conditions pooled	Their observed weights may differ from intended use
Only aggregate counts supplied	Cannot reconstruct individual paired flips
No statistical analysis supplied	Do not label slice gaps statistically established

Separate pictured facts and full reasoning

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

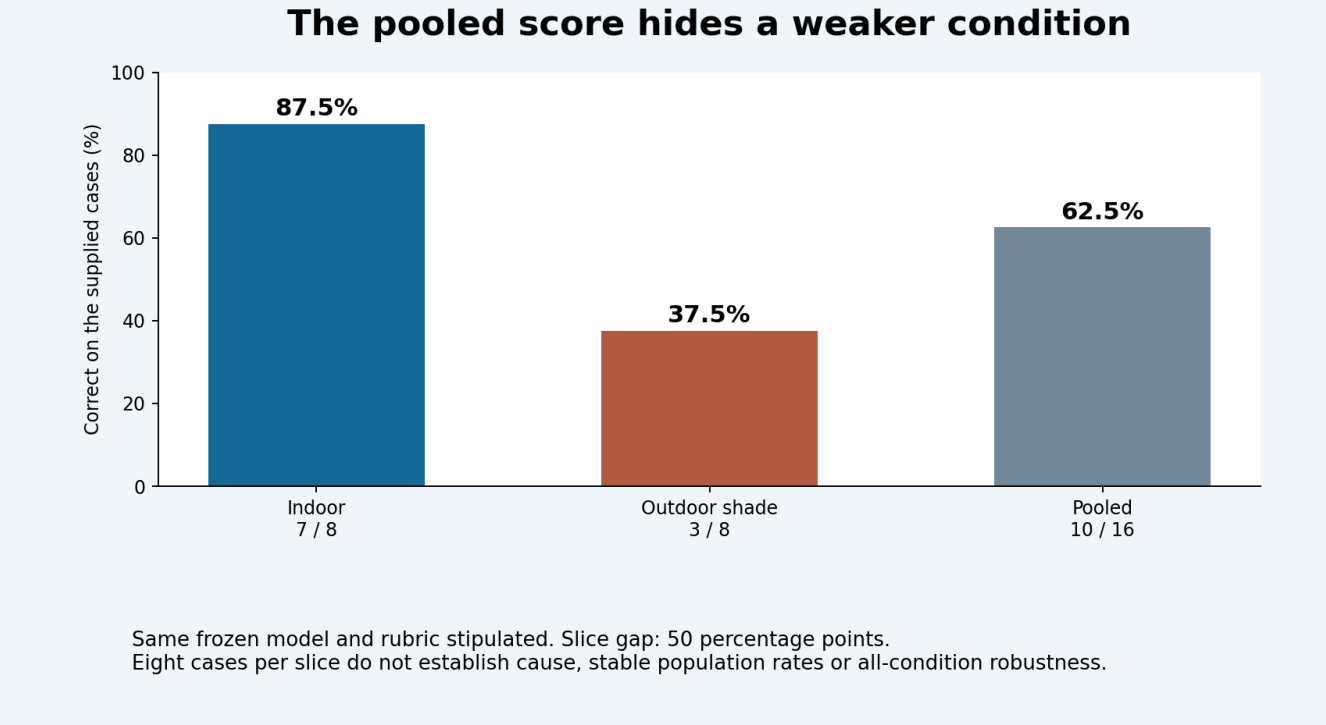
Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

The separate frozen-classifier picture has seven correct of eight indoor cases, or 87.5%, and three correct of eight shaded outdoor cases, or 37.5%. The descriptive slice gap is fifty percentage points. Combined correctness is ten of sixteen, or 62.5%. Reporting only that pooled score hides the weaker outdoor slice and is not a result for either setting alone.

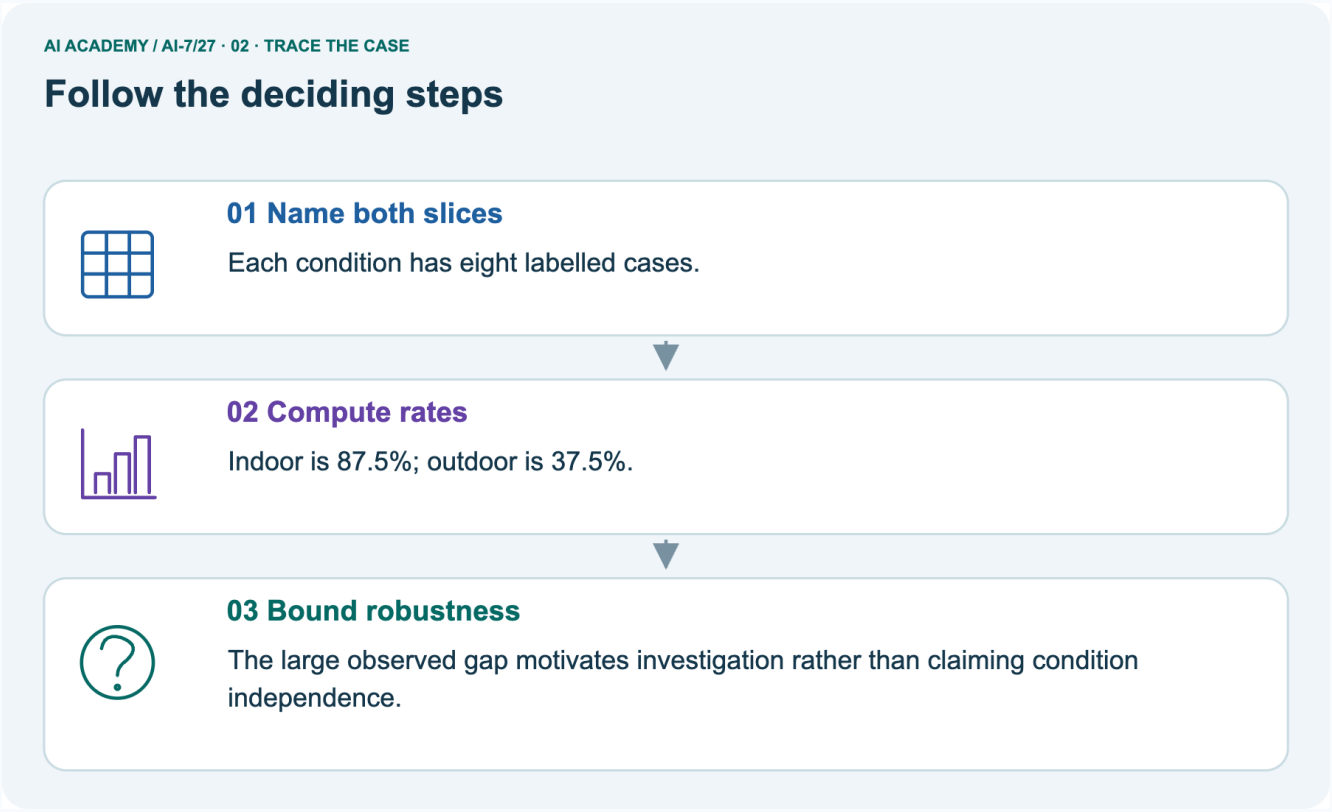
Eight cases per slice do not establish stable population rates or the reason for the difference. Keep the model and rubric frozen in the stated comparison, but do not mistake that exercise assumption for code-verified controls or full sampling evidence.

Separate picture slice	Correct / total	Observed rate
Indoor	7/8	87.5%

Separate picture slice	Correct / total	Observed rate
Outdoor shade	3/8	37.5%
Pooled	10/16	62.5%
Indoor minus outdoor	Different supplied slices	50 percentage points
Hidden by pooled-only report	Outdoor weakness	No separate outdoor rate visible
Why rates differ	Not supplied	No causal or significance claim



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name both slices	Each condition has eight labelled cases.
2. Compute rates	Indoor is 87.5%; outdoor is 37.5%.
3. Bound robustness	The large observed gap motivates investigation rather than claiming condition independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Eight cases per slice do not establish stable population rates or the reason for the difference.

Change: Report only the combined 10/16=62.5%.

Prediction: The arithmetic is correct, but the condition gap becomes hidden.

Reason: Combining records discards the slice distinction relevant to the deployment goal.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Robustness check	Evaluation under declared varied conditions
Condition-specific result	A score within one stated setting
Aggregate masking	A summary hiding differing condition outcomes

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Robustness across conditions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/27

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.

2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined 10/16=62.5%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Robustness is bounded by the tested variations and independent **evidence** policy.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

Each condition has eight labelled cases. Indoor is 87.5%; outdoor is 37.5%. The large observed gap motivates investigation rather than claiming condition independence.

Changed case: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Changed-case reasoning: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main current 9/10 = 90%, outdated 5/10 = 50%, gap forty percentage points. Do not claim the source condition alone caused the difference. Picture indoor 7/8 = 87.5%, outdoor 3/8 = 37.5%, gap fifty points; pooled 10/16 = 62.5% hides the weaker slice. Independent condition A baseline 6/8 = 75% and candidate 7/8 = 87.5% gives +12.5 points; condition B baseline 5/8 = 62.5% and candidate 4/8 = 50% gives −12.5 points.

Both pooled totals are 11/16, so the tie hides opposite condition effects. Per-case flip identities are not supplied despite paired-rate wording. No significance, universal advantage or causal attribution follows.

Diagnose the current misconceptions

The worked result establishes every future case: Robustness is bounded by the tested variations and independent **evidence** policy.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.

Teacher explanation: Candidate gains12.5 points inA but loses12.5 inB; overall11/16 for both. Aggregate equality hides condition differences.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Robustness check**s examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause. Keep baseline, metric and eligibility aligned across conditions.

Report condition counts and failures separately; new conditions may extend **evidence** but cannot guarantee all conditions.

W2 Trace the worked case

Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

W3 Audit the limit

A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Robustness is bounded by the tested variations and independent **evidence** policy.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Robustness is bounded by the tested variations and independent **evidence** policy.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims. Qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.

Teacher explanation: Candidate gains12.5 points inA but loses12.5 inB; overall11/16 for both. Aggregate equality hides condition differences.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Candidate gains12.5 points inA but loses12.5 inB; overall11/16 for both. Aggregate equality hides condition differences. Require intermediate reasoning and the stated scope.

Fresh retry: A baseline4/6, candidate5/6; B baseline5/6, candidate4/6. Expected reasoning: Gains/losses16.667 points; overall9/12 each. Preserve condition-specific findings.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled robustness across conditions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of robustness across conditions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Negative results and claim revision. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/28, ai-7/30, ai-7/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of robustness across conditions

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

Check every learner before discussion: Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Robustness is bounded by the tested variations and independent **evidence** policy.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Report only the combined 10/16=62.5%. Require a **prediction** and a reason before comparing. Expected result: The arithmetic is correct, but the condition gap becomes hidden.

Reason: Combining records discards the slice distinction relevant to the deployment goal.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

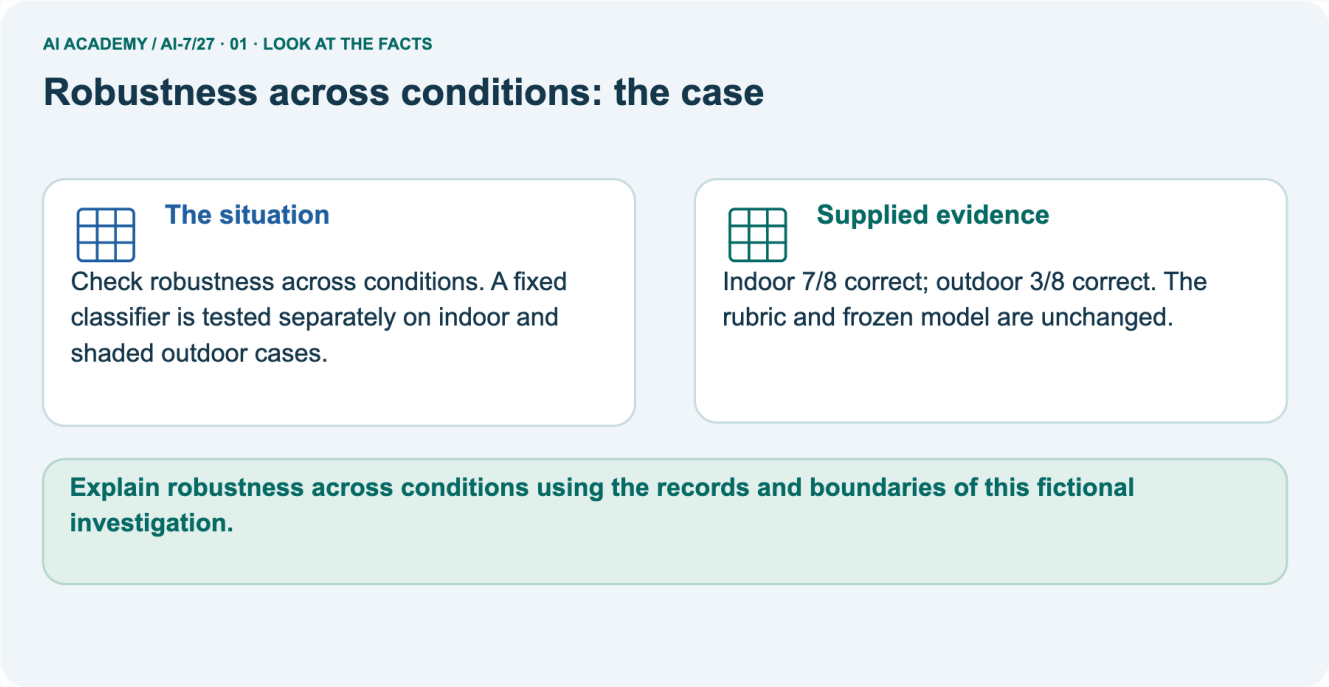
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Robustness across conditions: the case



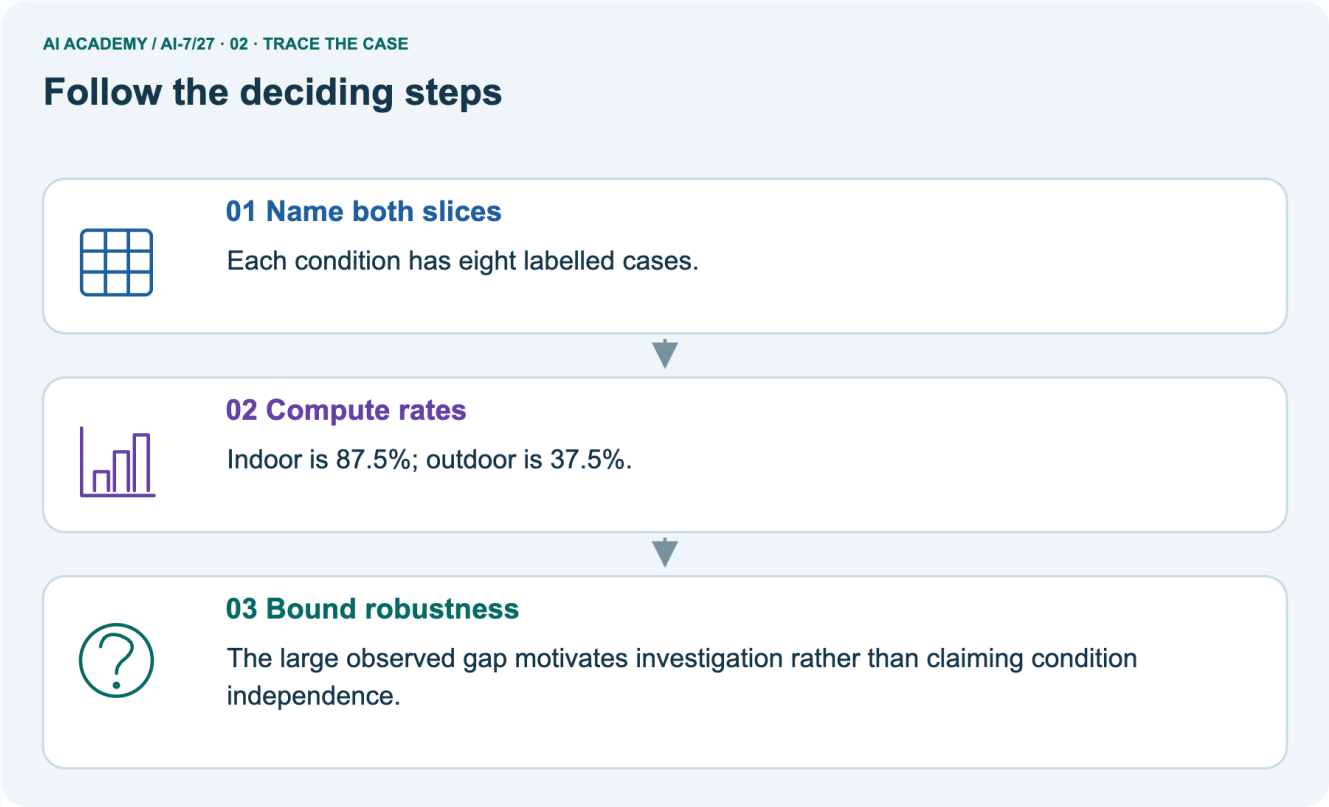
Follow the labelled picture with the matching explanation. Open the vector diagram

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Supplied facts: Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Name both slices: Each condition has eight labelled cases. → Compute rates: Indoor is 87.5%; outdoor is 37.5%. → Bound robustness: The large observed gap motivates investigation rather than claiming condition independence.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Fixed model across conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

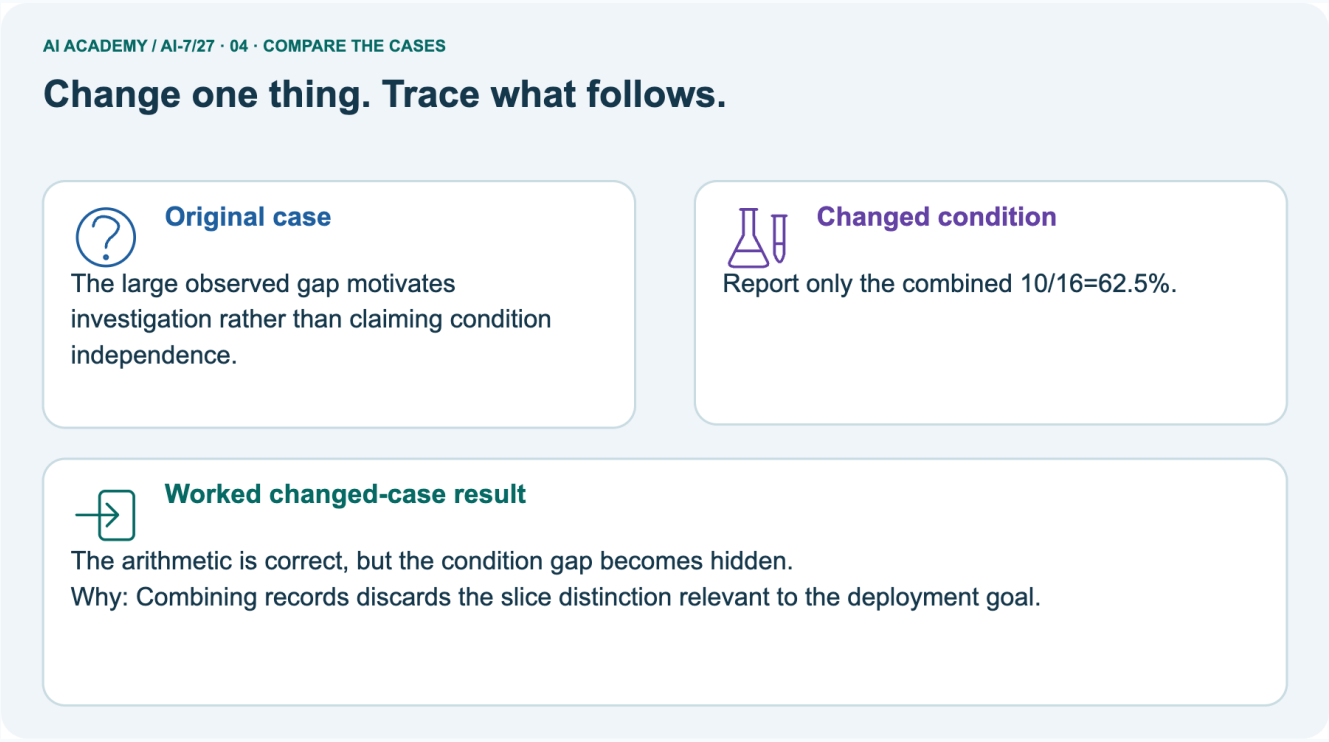
Fixed model across conditions. Units: correct cases out of 8 per condition. Indoor: 7. Shaded outdoor: 3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The large observed gap motivates investigation rather than claiming condition independence. Changed condition: Report only the combined 10/16=62.5%.

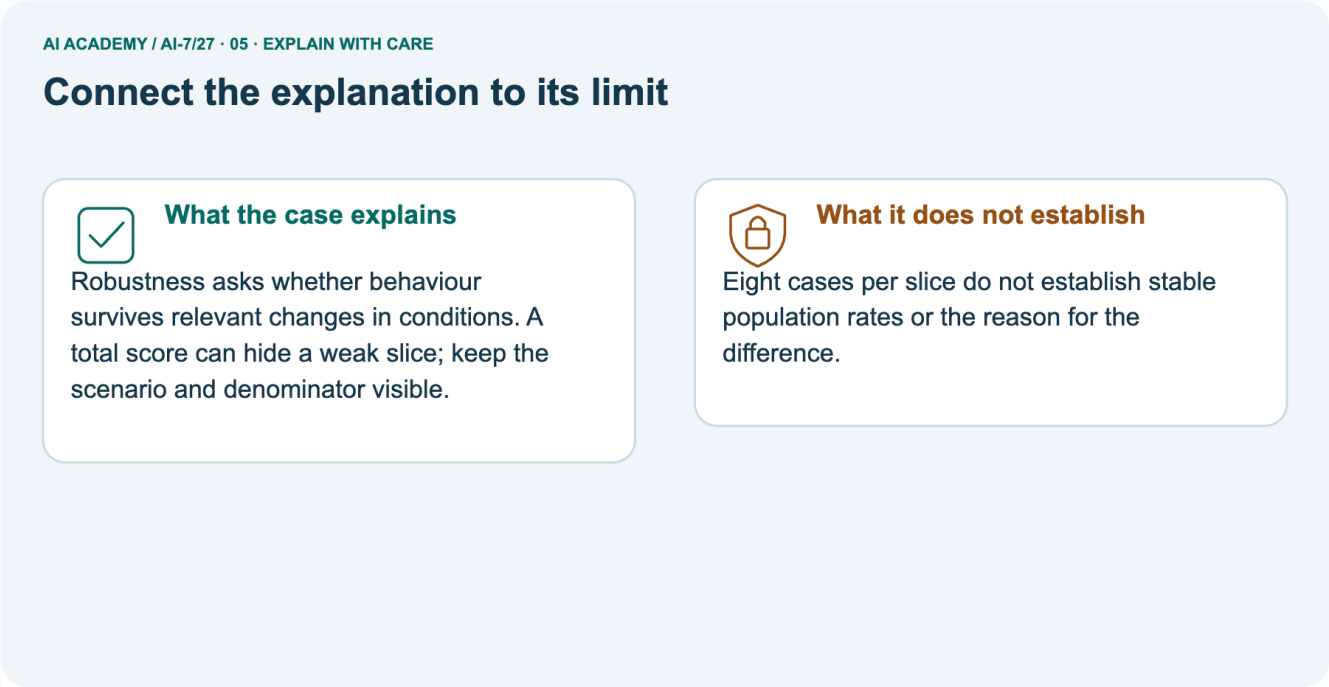
Expected result: The arithmetic is correct, but the condition gap becomes hidden. Reason: Combining records discards the slice distinction relevant to the deployment goal.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Combining records discards the slice distinction relevant to the deployment goal.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

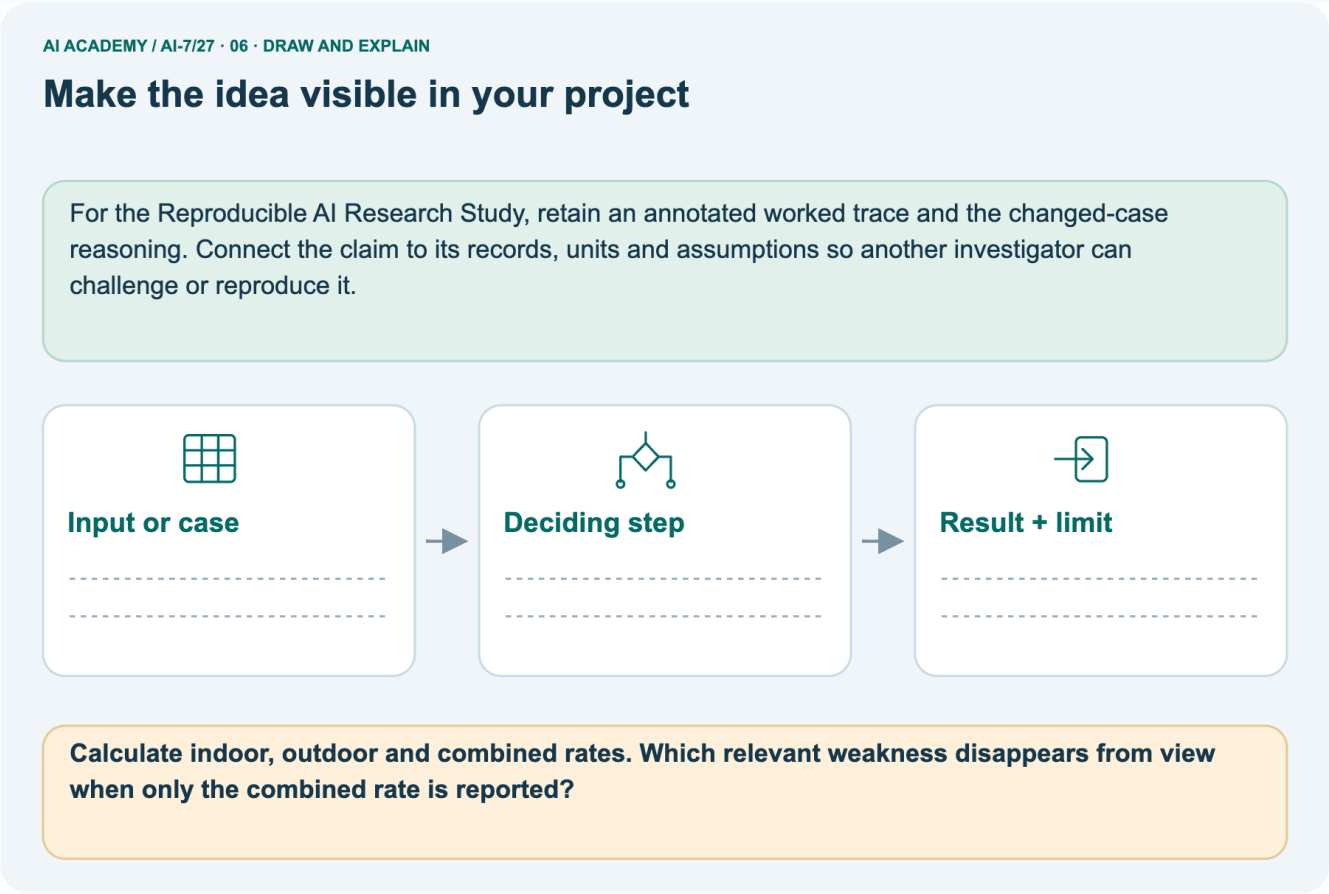
Limit: Eight cases per slice do not establish stable population rates or the reason for the difference.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Eight cases per slice do not establish stable population rates or the reason for the difference.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Each condition has eight labelled cases. Indoor is 87.5%; outdoor is 37.5%. The large observed gap motivates investigation rather than claiming condition independence.

Changed case: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Robustness across conditions: the case

02. Follow the deciding steps

03. Fixed model across conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.